

No online multiplayer for GRID

Codemasters has shut down online gameplay of its GRID franchise, as a third party didn't renew its contract to host the servers.

Kinect SDK for Windows out

Microsoft has released Kinect SDK for Windows developers. It's still in beta, free to download and requires Windows 7.

HOW WE TESTED

We divided the routers into 3 types:

1) ADSL routers

2) Regular routers

3) Routers with 3G supporting USB ports

For the purposes of testing, each router was connected to one notebook via a wired connection, and to the other notebook via its wireless adapter. File transfers, media streaming and such was done from the notebook connected wirelessly. All media and data files were placed on the primary laptop connected to the routers via LAN. This way, we eliminated any bottlenecks from the router to the data, thereby testing only the connection between the router and the notebook connected wirelessly. Obviously, the adapter on the secondary notebook would also be a bottleneck, but there is no way around this. Also remember that we measure our scores in megabytes per second and not megabits. The 802.11n rating specifies 300 mbps, or 300 megabits per second. When we measure a score of 5 MBps (megabytes per second), we mean 5 x 8 or 40 mbps, since one byte equals eight bits.

The primary notebook consisted of a Core i5 480M CPU, 4 GB of RAM, and a 500 GB hard drive via a CAT 6 cable. The secondary (wirelessly connected) notebook had a Core i5 480M, 3 GB of RAM, and a 320 GB hard drive. Each router was reset to its factory settings before testing. We disabled DHCP on each of the routers and assigned each notebook adapter a static IP, which we changed to match the default IP range of the router. Then we disabled wireless security. We forced the routers to 802.11n, except for those that didn't support the protocol. If a transfer rate was mentioned, we set it to its maximum value. Wireless signal strength was set to 100 per cent, and if the router supported dual simultaneous band, this feature was enabled with the same network SSID. In Windows 7, we used the "home network" setting, to enable data and media sharing.

As usual the secondary notebook was moved through three zones, in increasing magnitude of signal attenuation. Zone 1 was a snap, with the notebook placed within 6 feet of the router and an 80 per cent line of sight. For zone 2 we moved the notebook so as to get a single thin cement pillar and 3 wooden partitions between the router and the secondary notebook. For zone 3, we moved the notebook to the first floor, (the router was one storey below), with the same lateral gap between them as in zone 2, although now there was an entire cement + steel roof with mezzanine that was at least 2 feet thick. Thus zone 3 was the most stringent of our zones, and naturally we assigned greater weights to scores in this zone. The approximate distance between router and notebook in zone two was 20 feet, and in zone three was 30 feet.

Our benchmarks consisted of Performance Mark 7's network test, whereby data is streamed between the two notebooks. Obviously, the wireless network is a lot slower than the LAN connection, and the network adapter on the secondary laptop is constant, so the difference in speeds can be purely attributed to the different routers used. For our real world tests, we transferred data between both notebooks. The sequential test involves a single 50 MB .RAR file, whereas the non-sequential test consists of multiple small files, of varying sizes and types that total up to 50 MB. Finally, we streamed a movie wirelessly in each of the three zones and assigned a subjective score on 10 based on how each router performed. 8.75 and above out of 10 meant there was basically no issue with playback, even with play/pausing, and skipping through the movie file. We tried skipping forward and backward. A score of 7.5 to 8.5 meant there was problem during skipping with some minor audio sync issues and video taking a second or so to resume after a skip. A score of below 7.5 meant the video took up to 3 seconds to resume with some audio clipping, while a score below 5 meant that the experience was quite poor. Routers that failed to play the file or lost connection to the wireless network got a 0.

throughput in zone two over zone one, but in zone three it lost signal completely and wasn't able to complete the Performance Mark 7 test. It also failed our real world transfer. Its sibling, the E1000 was slower in zone one and two, but managed to complete the test in zone three as well. Incidentally, the Linksys E1000 performed flawlessly at streaming video in zone one. The most surprise performer, however, turned out to be Netgear's older 802.11g supporting WGR614. With good scores throughout zones, it aced the video streaming tests, shaming many faster and costlier routers.

Despite its decent looks and rather large antenna, the Buffalo WCR-HP-GN performed poorly. It failed the file transfer in zone three, and even in the other two zones, speeds were snail-like. In stark comparison, the tiny Belkin Basic was quite a speed demon. With speeds of 3.6 MBps in the

sequential transfer in zone three and 3.27 MBps in the non-sequential test the Belkin Basic shames a lot of the other costlier routers. The only place it lost out was in the movie tests – it did well, but when skipping through the video, the audio dropped for a couple of seconds before kicking in.

If you're shopping for a basic router it's hard to look past the Belkin Basic – for the price of ₹2,399, the performance was anything but basic – it outperformed some routers costing twice as much. It wins our Best Buy award, a unanimous decision that its performance numbers amply support.

BETWEEN ₹2,501 AND ₹5,000: LESS FOR MORE?

This category consisted of nine routers, three from SMC, two from Belkin and one each from ASUS, Buffalo, Netgear and ZyXEL. Most of these routers

featured USB ports, some of them for exclusively for use with 3G data cards, the others for media sharing. Routers with the media sharing feature are interesting options, for any media attached to them can be accessed like a shared device over the network, sort of like a NAS drive. Any media files added to the public folder of the drive can be streamed wirelessly to a PC or a notebook. Content can also be blocked, and most routers will also allow sharing of this drive online, as an FTP. Generally, drives with USB ports used to be priced over the 5,000-Rupee mark, but as with all things computer, they've become more affordable.

Looks and features

The Belkin Surf and Share look similar, but the Surf is identical in size to the Basic, while the Share is slightly larger router, similar to the Belkin Play. Other than that,

the Share has a single USB port. Within the routers firmware, another difference becomes visible – the Share is faster than the Surf, and while the Surf takes up to 20 seconds to apply a setting, whereas the Share takes only 6 seconds. The Netgear WNR2200 looks smart. Its got a nice stand that integrates well into the router giving the appearance of a single piece, and the router can be wall mounted as well as being laid flat on a table. There are dedicated buttons to turn off wireless radio and enable WPS. It has a single USB port that can be used for media and works with a thumb drive as well as an external HDD.

The ASUS WL-330N3G was the tiniest router in the entire test. In fact, it can fit into your pocket. The single USB port is meant only for 3G data cards and doesn't support hard drives. The SMC SMCWBR14S-3GN looks identical to the SMCWBR14S-